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Towards a Constrained Understanding of Europa's Subsurface Driven by Magnetic Induction Sounding

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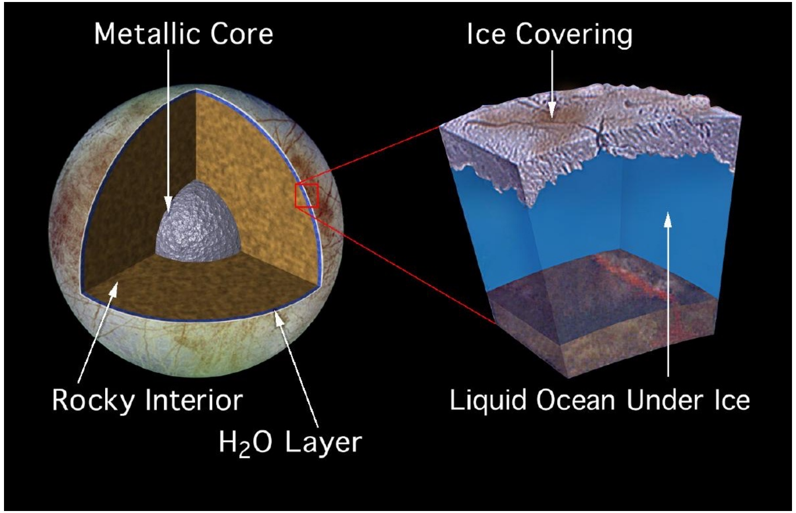
Europa: Background

Europa: $R_E = 1561$ km



NASA/JPL

- Possesses tenuous, O_2 rich atmosphere
- Embedded within Jupiter's magnetospheric / plasma environment
- Target of the upcoming JUICE and Europa Clipper missions
- Favorable candidate for habitable qualities (tremendous water content)

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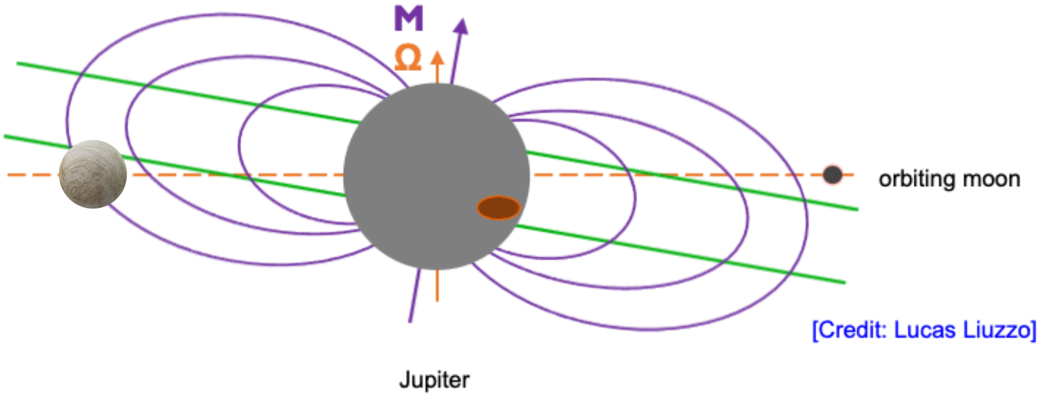
Europa: Background

M: Magnetic Axis

Ω: Rotation Axis

Plasma Sheet



Analogs:

- Offset between Jupiter’s spin and rotation axes
- ➔ time-varying magnetospheric field **B** at Europa’s orbit
- Drives induction in conducting layers of the moon (salty water)
- Led to the discovery of Europa’s ocean! (Kivelson et al., 2000)

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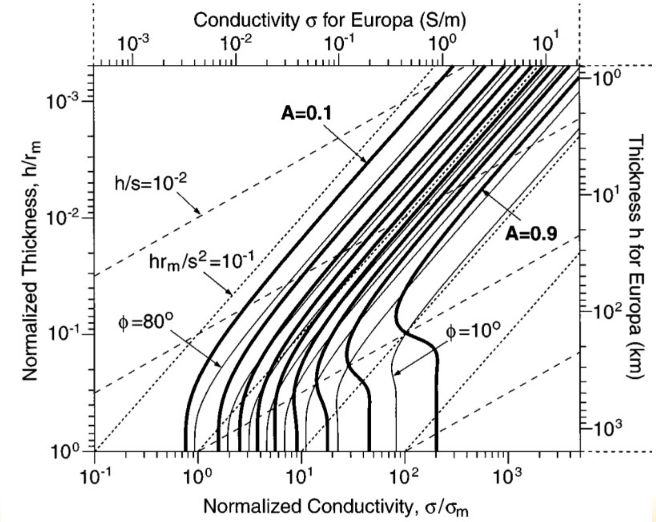
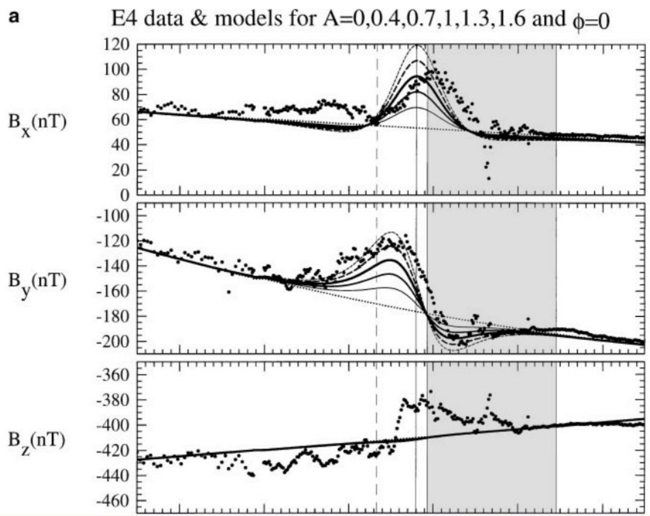
Science Question

- However, even today Europa's subsurface stratification remains poorly constrained
- Studies have applied various methods to obtain the depth/conductivity profile
- Often the nature of Europa's local environment is not considered, and the choice of inversion strategy is inspection or L-S (e.g., Weber et al., 2022)





Zimmer et al. (2000)

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Our goal: tighter constraints on Europa's subsurface conductivity profile

To do so, we use magnetometer data from Europa's environment, and:

- more robust inversions (truncated SVD, nonlinear methods)
- highly selective pre-processing data analysis routine (background prevention & subtraction)

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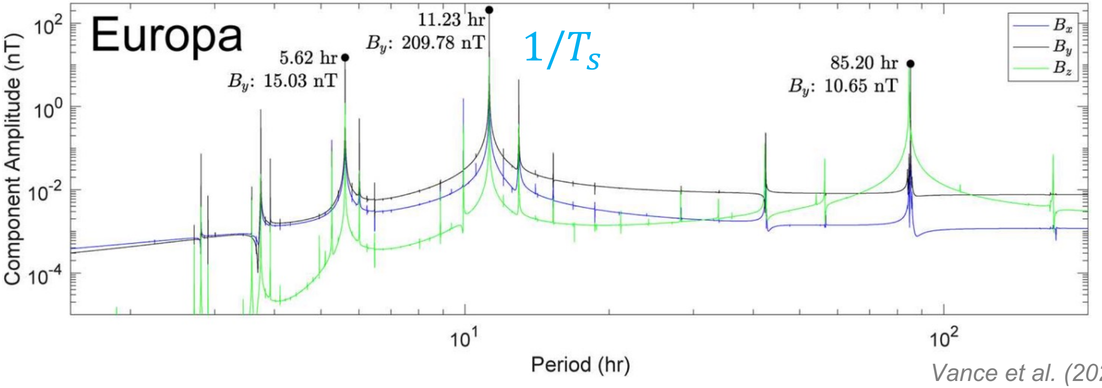
Forward Operator: Induction Physics

- Dominant harmonic: synodic rotation period of Jupiter
- Other inducing harmonics



Vance et al. (2021)

- Induction described by:

$$\nabla^2 \mathbf{B} = \mu_0 \sigma \, \partial_t \mathbf{B} \quad ; \quad \text{if } \sigma \longrightarrow 0 \quad \Longrightarrow \quad \nabla^2 \mathbf{B} = 0$$

➔ spherical harmonic decomposition assuming spherical symmetry isotropic shape

- For a prescribed subsurface conductivity profile, the inductive response is:

$$\mathbf{B}_{\text{ind}} = \frac{\mu_0}{4\pi} \left(3(\mathbf{r} \cdot \mathbf{M}_{\text{ind}})\mathbf{r} - \mathbf{r}^2 \mathbf{M}_{\text{ind}} \right) / r^5$$

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Forward Operator: Induction Physics

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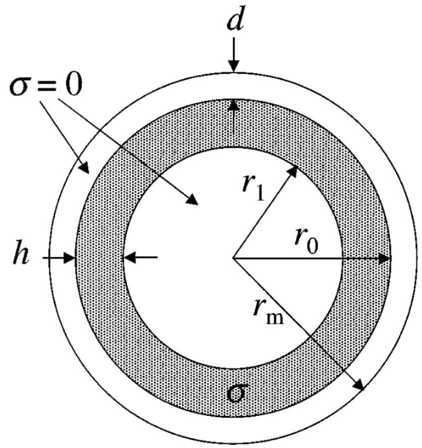
$$\mathbf{M}_{\mathrm{ind}} = -A\,e^{-i(\omega t-\phi)}\,\frac{2\pi R_E^3}{\mu_0}\,\mathbf{B}_{h,0}$$

(Zimmer et al., 2000)



➔ A, ϕ are amplitude and phase lag of induced field signal

➔ Assume that Europa’s subsurface is discretized into 3 layers:

➔ A, ϕ dependent on subsurface ocean thickness, depth, σ

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Forward Operator: Induction Physics

$$\mathbf{M}_{\text{ind}} = -A\,e^{-i(\omega t-\phi)}\,\frac{2\pi R_E^3}{\mu_0}\,\mathbf{B}_{h,0}$$

$$Ae^{i\phi} = \left(\frac{r_0}{R_E}\right)^3 \frac{C J_{\frac{5}{2}}(r_0\,k) - J_{-\frac{5}{2}}(r_0\,k)}{C J_{\frac{1}{2}}(r_0\,k) - J_{-\frac{1}{2}}(r_0\,k)}$$

Amplitude, phase lag dependence

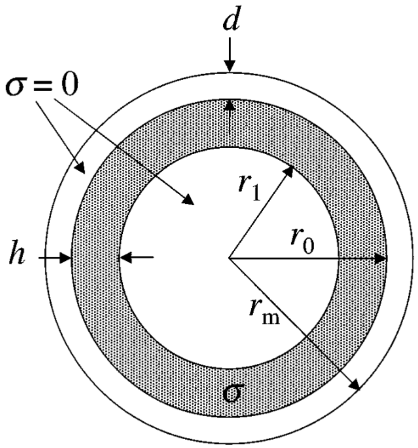
$$C \equiv \frac{r_1 k J_{-\frac{5}{2}}(r_1\,k)}{3J_{\frac{3}{2}}(r_1\,k) - r_1 k J_{\frac{1}{2}}(r_1\,k)}$$

coefficient dependent on ocean depth

$$k = (1-i)\sqrt{\frac{\mu_0\sigma_o\,\omega}{2}} = (1-i)\sqrt{\frac{\pi\mu_0\sigma_o}{T_s}}$$

complex wavenumber dependent on conductivity value

(Zimmer et al., 2000)



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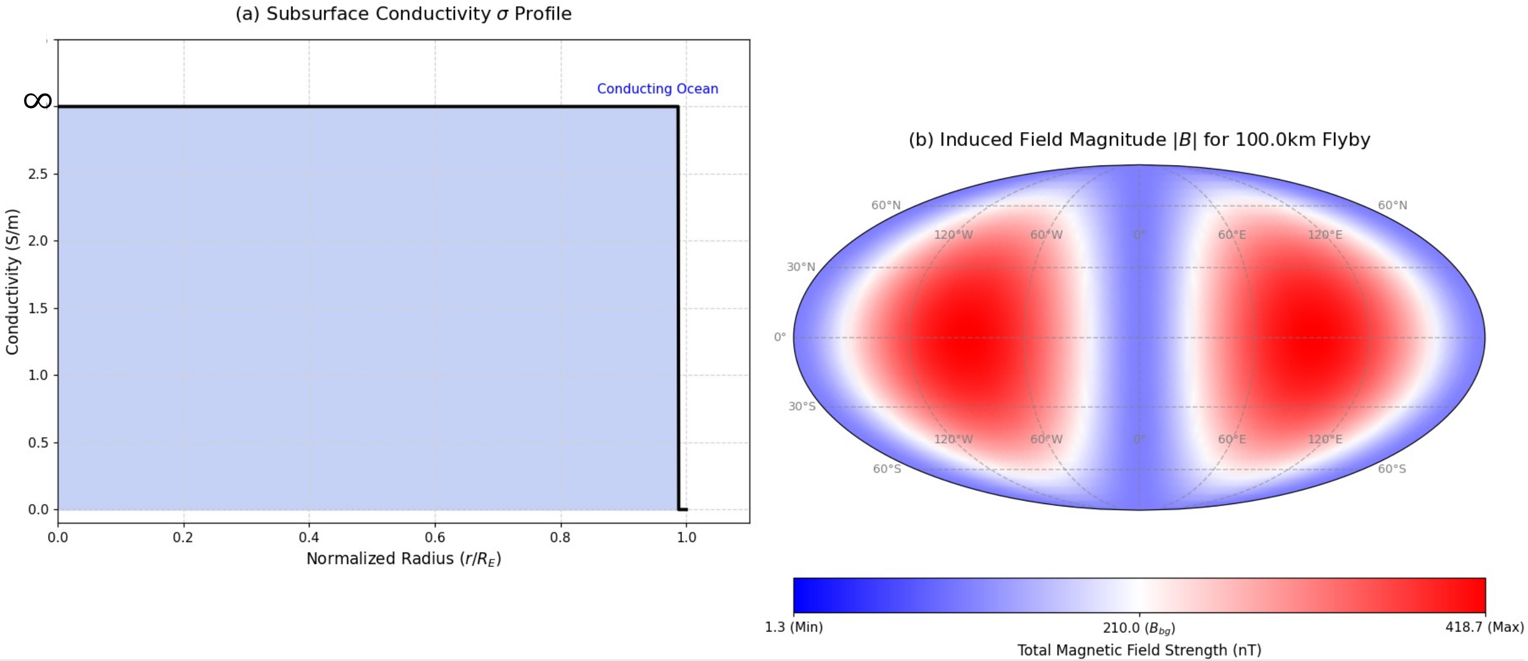
Inductive Response: Perfect Conductor

- In the most coarse treatment, Europa is considered a perfect spherical conductor



➔ $A \to 1, \phi \to 0^{\circ}$

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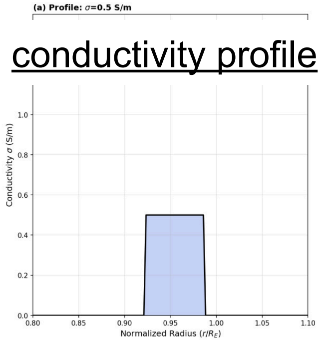
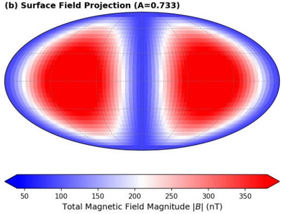
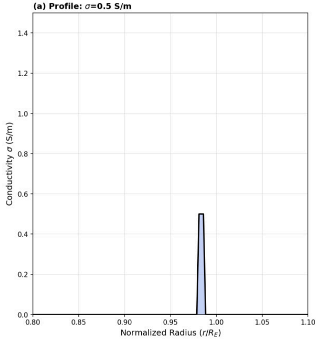
Inductive Response: 3-Layer Approximation

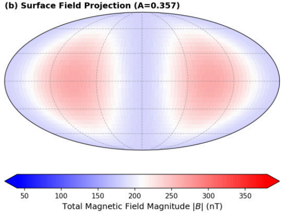
conductivity profile | surface $|\mathbf{B}|$

1


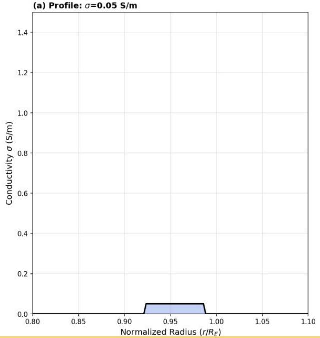


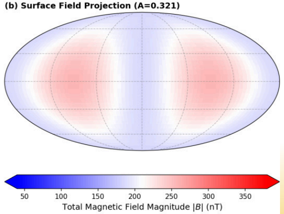
100 km thickness ➔ $A \to 0.73$

2




10 km thickness ➔ $A \to 0.36$

3




100 km thickness, weak σ ➔ $A \to 0.32$

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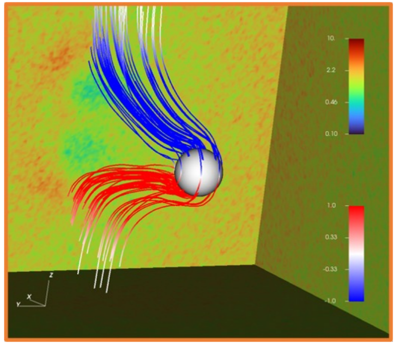
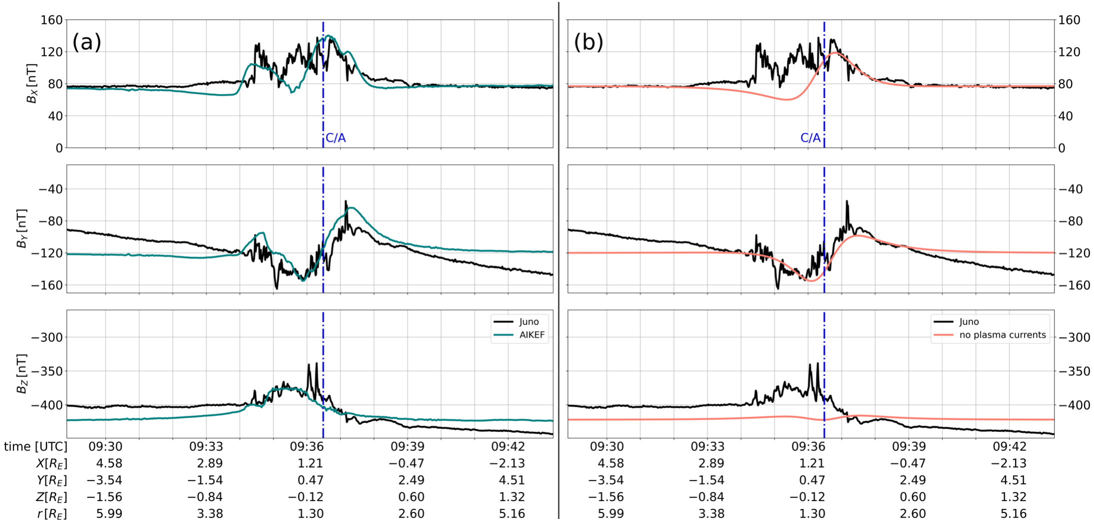
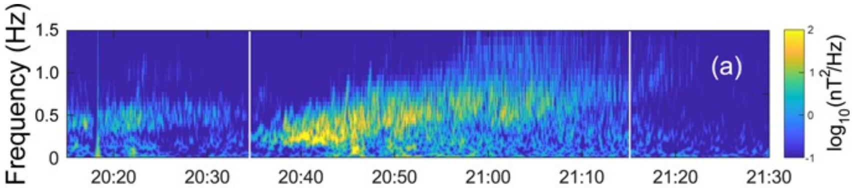
Contaminating magnetic “Noise”

- Europa possesses a rich, dynamic plasma interaction with Jupiter’s magnetosphere:
- Pile-up, draping, Alfvén Wings:





- Plasma wave modes:



(Addison, Haynes et al., 2024)

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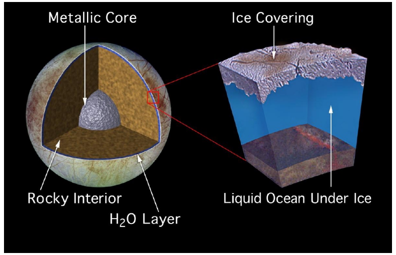
Summary: Magnetic Induction Sounding at Europa

<u>Goal</u>: Constrain the subsurface stratification of Europa



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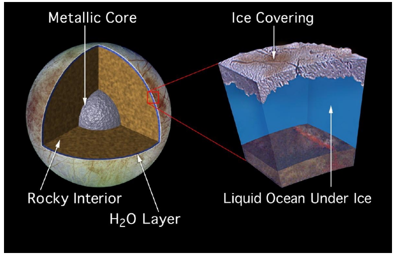

Method: Combine in-situ magnetometer observations, context-driven data filtration and assumption of a salty, symmetric liquid layer

$$\mathbf{B}_{\mathrm{ind}} = \frac{\mu_0}{4\pi}\left(3(\mathbf{r}\cdot\mathbf{M}_{\mathrm{ind}})\mathbf{r} - \mathbf{r}^2\mathbf{M}_{\mathrm{ind}}\right)/r^5$$

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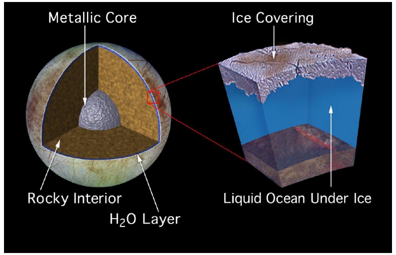

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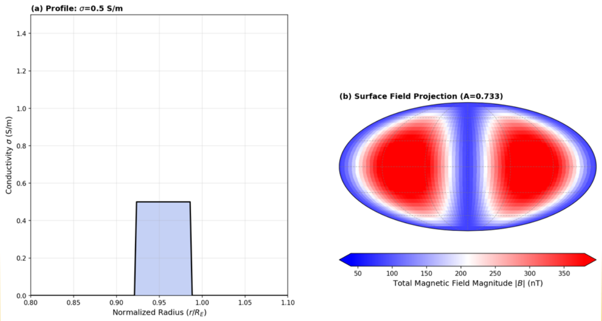
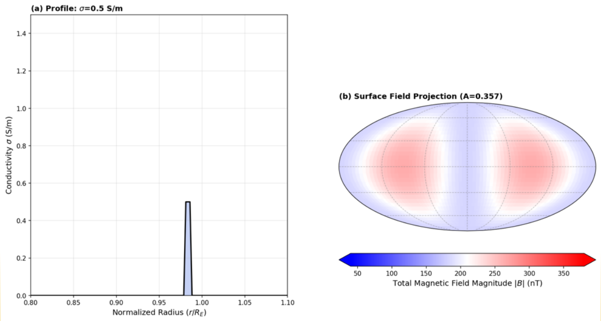
Results: Improved constraints on the depth, thickness, and bulk conductivity of Europa's subsurface liquid water ocean





Thank you